

CA1 - Lesson 01

Electric Circuits

SCHOOL OF
ENGINEERING

E117

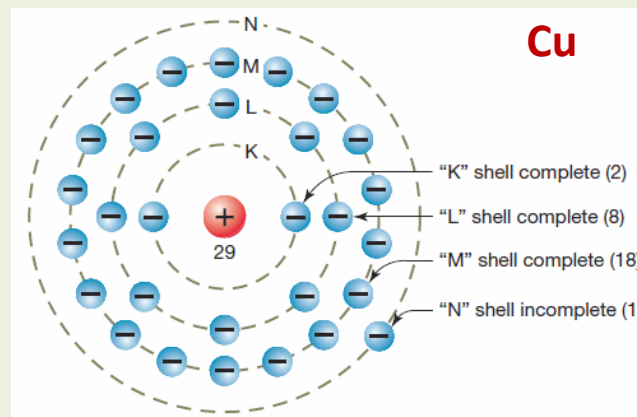
Principles of
Electrical and
Electronic
Engineering



Refer to Smartbook Sec 2.1 Atoms and Ions

- When **atom** contains an **equal number of electrons and protons**, its combined **electric charge zero**.
- The outermost shell of an atom is called the **valence shell** and its electrons are called valence electrons.
- The energy level is the highest at the valence shell. With enough **external force**, an atom may lose or acquire electrons.

A **negatively charged ion** is an atom that has **acquired** electrons.



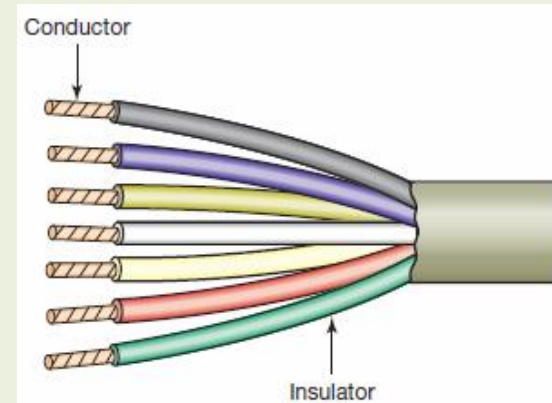
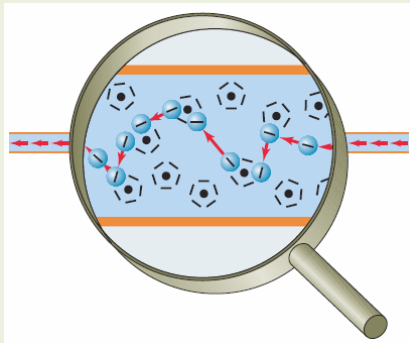
A **positively charged ion** is an atom that has **lost** electrons.

- The process by which atoms either gain or lose electrons is call **ionization**



Refer to Smartbook Sec 2.4 Electricity Defined

- The flow of electrons through a conductor is defined as **Electricity**
- **Conductors** offer very little resistance to the flow of electrons through them
- **Insulators** have few, if any, free electrons and resist the flow of electrons



Question: Name 1 common Conductor and Insulator material

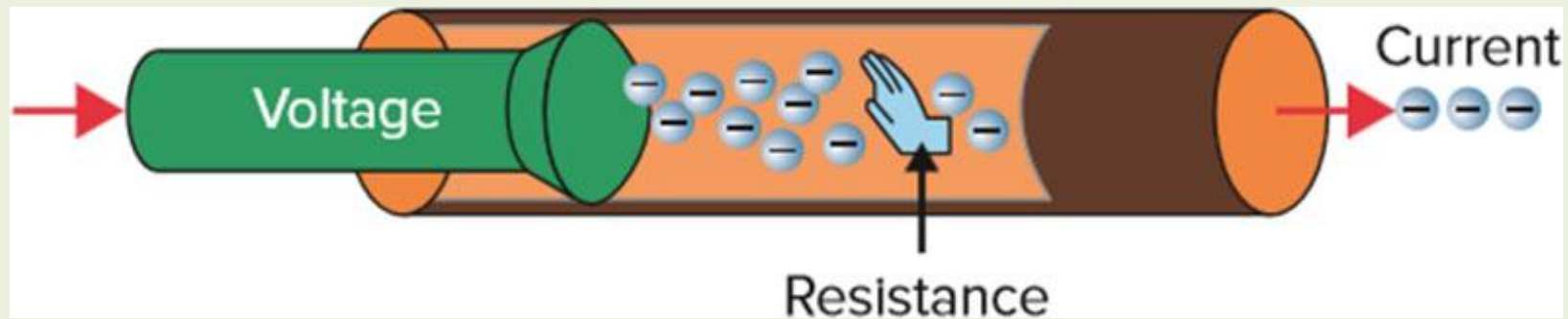
Copper wire; rubber shoe



Refer to Smartbook Sec 4.1 Current, Voltage and Resistance

Voltage (SI unit: V) is the measure of electrical pressure, Electromotive Force (EMF) that causes current flow.

Current (SI Unit: A) is a measure of the **rate of electron flow** through a material.

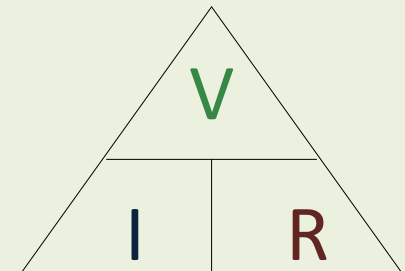


Resistance (SI Unit: Ω) is a measure of the opposition to the electron flow

Voltage, $V = I \times R$

Current, $I = V/R$

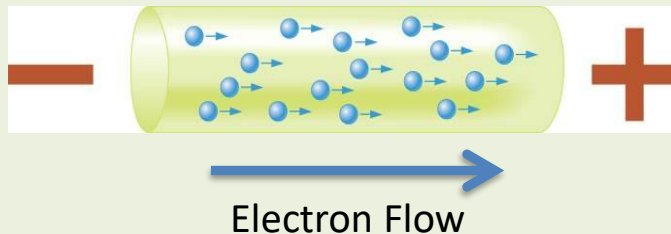
Resistance, $R = V/I$





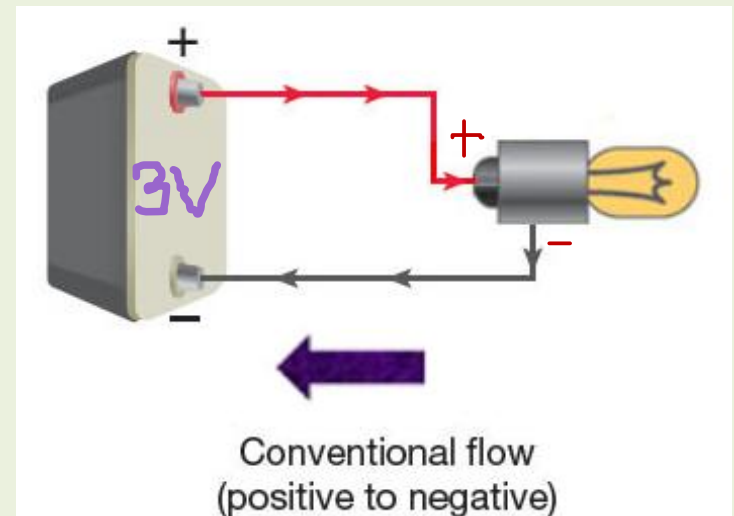
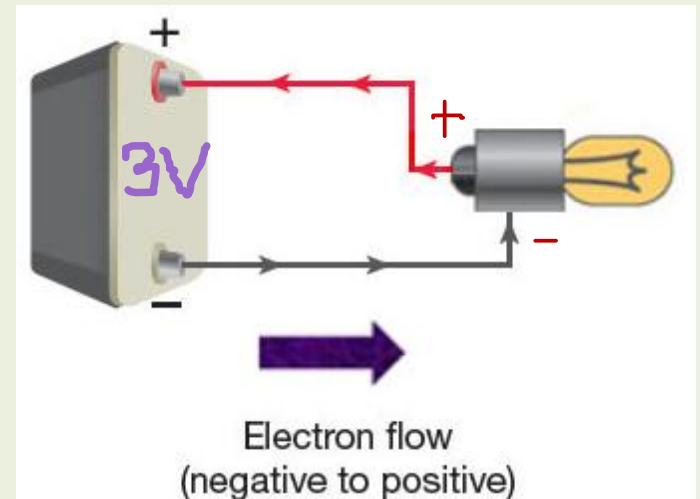
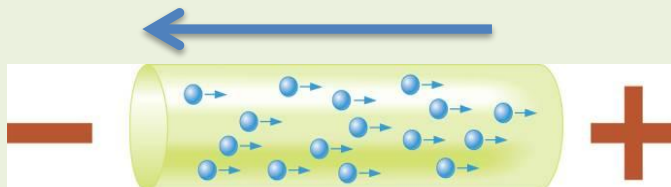
Refer to Smartbook Sec 4.8 Direction of Current Flow

Electron flow is based on the electron theory of matter and follows the motion of electrons in the circuit from **negative to positive**.



Conventional current flow is based on an older theory of electricity and assumes a current flow in the **opposite** direction from **positive to negative**.

Conventional Current Flow Direction





Refer to Smartbook Sec 4.6 Electric Circuit

If the circuit is broken at any point (**open circuit**), there is no longer a closed loop, and no current can flow.

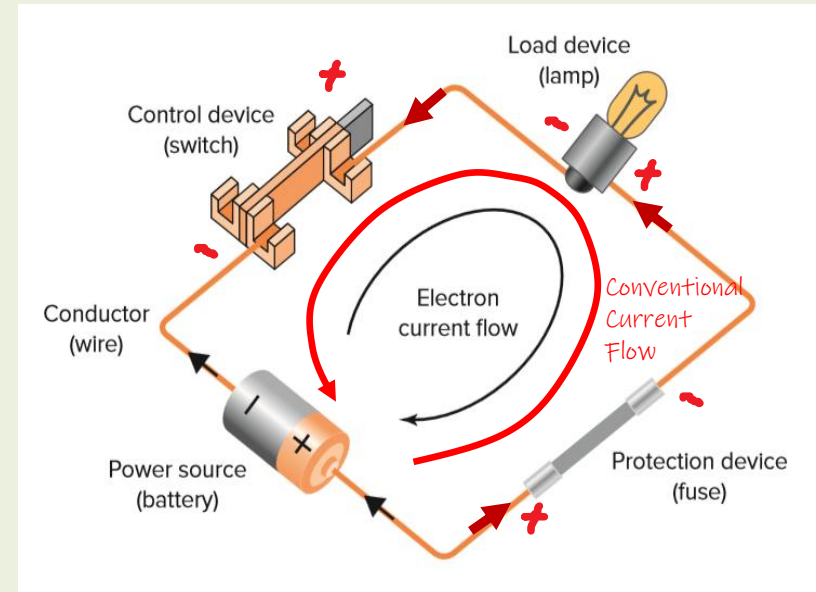
Source - provides the **voltage** and **current** required to operate the load.

- Conventional current is **leaving** at the positive terminal of a source

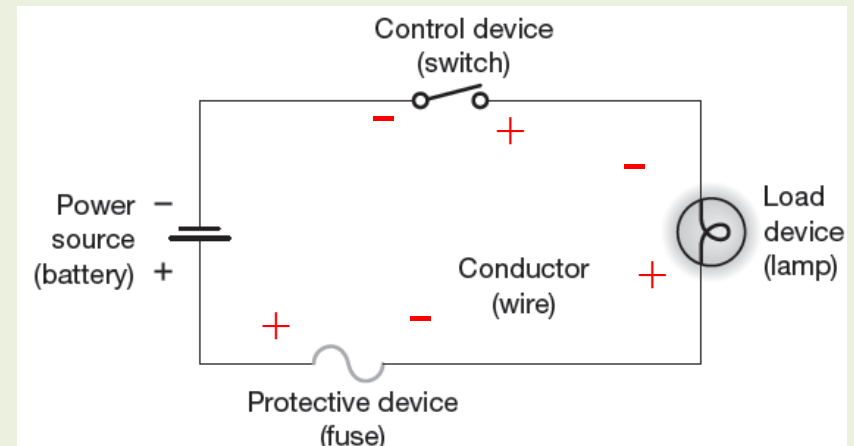
Load - is the component that **consumes electric power** rather than provides it.

- Conventional current is **entering** at the positive terminal of a load

Question: Label the polarity of each component and the conventional current direction in the schematic



A closed circuit

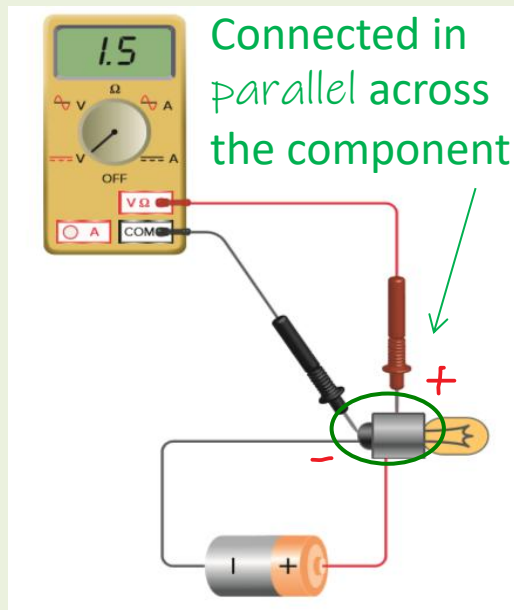


Equivalent Circuit Schematic diagram

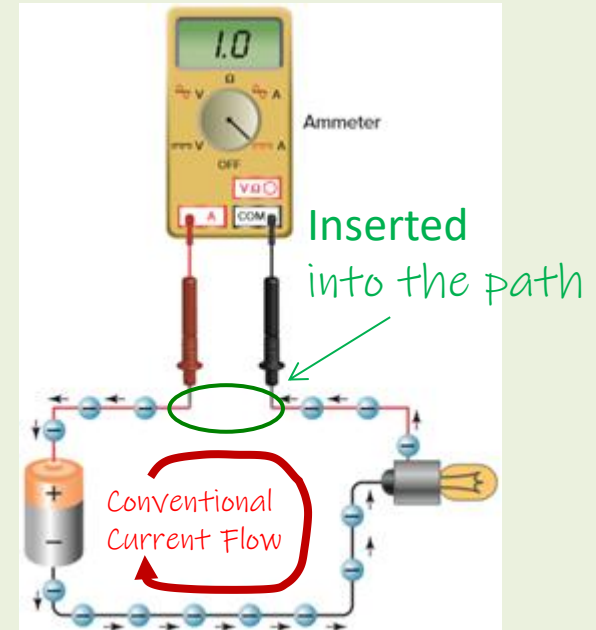


Refer to Smartbook Chapter 6.3/4 Measuring Voltage/Current

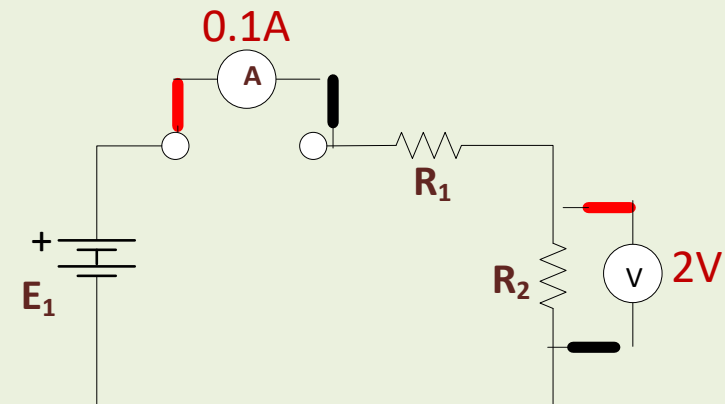
Voltmeter - to measure **potential difference** across **2** points in a circuit.



Ammeter – to measure **current** in a circuit.



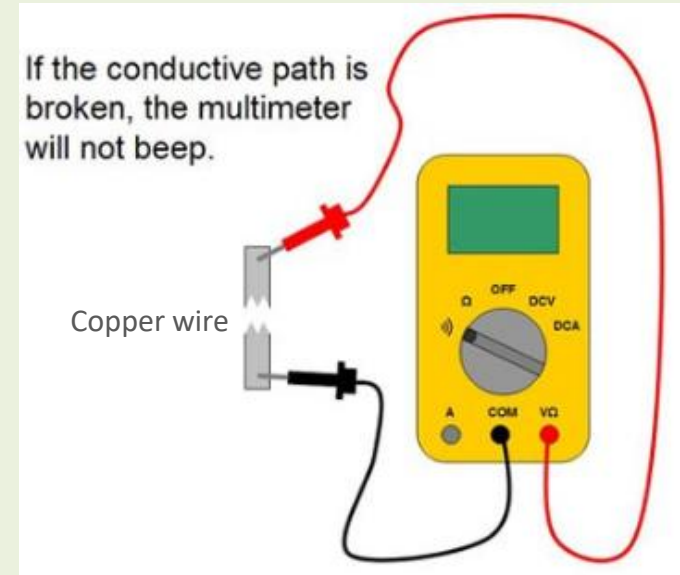
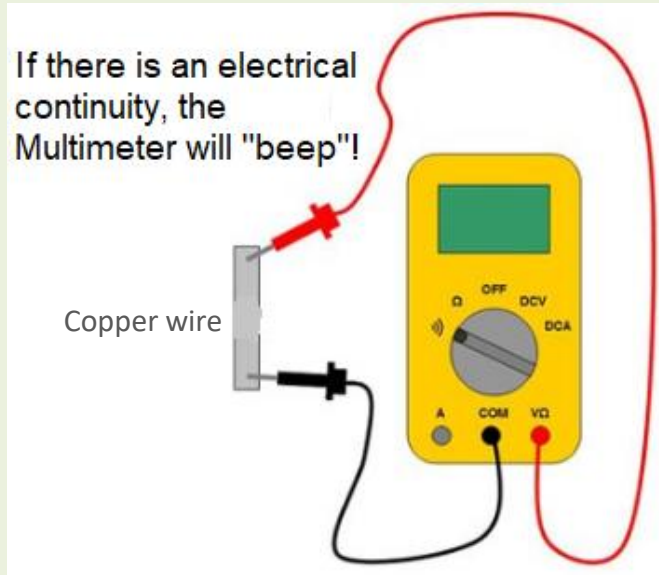
Question: Place the appropriate color leads onto the voltmeter and ammeter to measure positive voltage and current





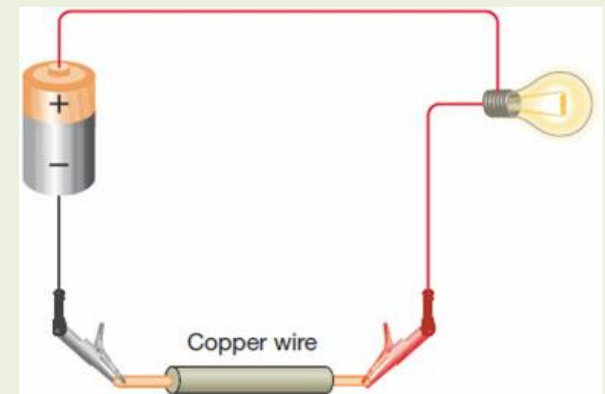
Refer to Smartbook Chapter 6.5 Measure Resistance and continuity test

When an electric circuit is capable of **conducting current**, it demonstrates **electrical continuity**.



Question: Does it matter if the color leads are swapped when testing continuity?

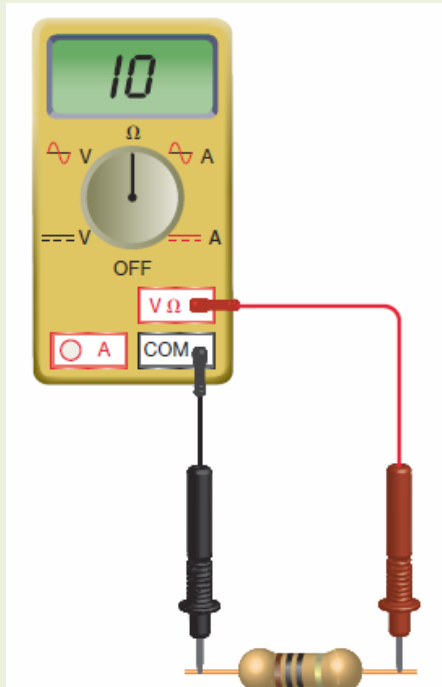
NO, the wire itself has no current flowing through





Refer to Smartbook Chapter 6.5 Measure Resistance and continuity test

- An **ohmmeter** is used to measure resistance in **Ohm (Ω)**
- Resistors are commonly used to adjust or set voltage and current level in a circuit
- Through hole resistor has color code used to identify the resistance value and tolerance



Ohmmeters should never be connected to live circuits!

0 1 2 3 4 5 6 7 8 9 0 1 2 3 4 5 6 7 8 9 Black Brown Red Orange Yellow Green Blue Violet Grey White ±1% ±2% ±5% ±10% Brown Red Gold Silver Color Codes	±1% ±2% ±5% ±10% 27K EXAMPLE 0 1 X1 1 1 X10 2 2 X100 3 3 X1000 4 4 X10000 5 5 X100000 6 6 X1000000 7 7 ±10 8 8 ±100 9 9 4 Band Resistors	±1% ±2% ±5% ±10% 15K EXAMPLE 0 0 X1 1 1 1 X10 2 2 2 X100 3 3 3 X1000 4 4 4 X10000 5 5 5 ±10 6 6 6 ±100 7 7 7 7 8 8 8 8 9 9 9 5 Band Resistors
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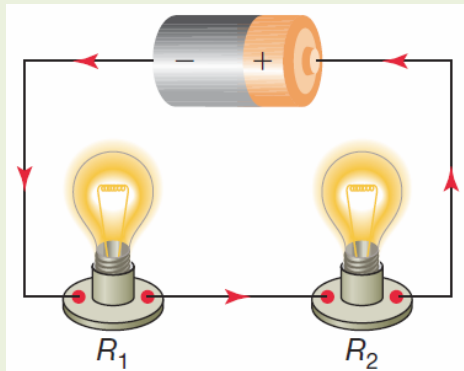


Refer to Smartbook Sec 11.1 Series Circuit Connection

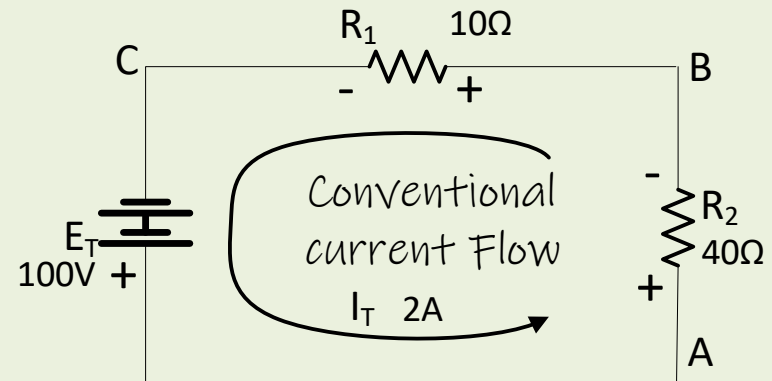
A **series circuit** may contain one or more loads but only **one path** for the current to flow.

If the current path is **broken** at any point all current flow in the circuit **stops**.

Conventional current **enters the positive terminal and exits from the negative terminal of a load**.



Electron Flow



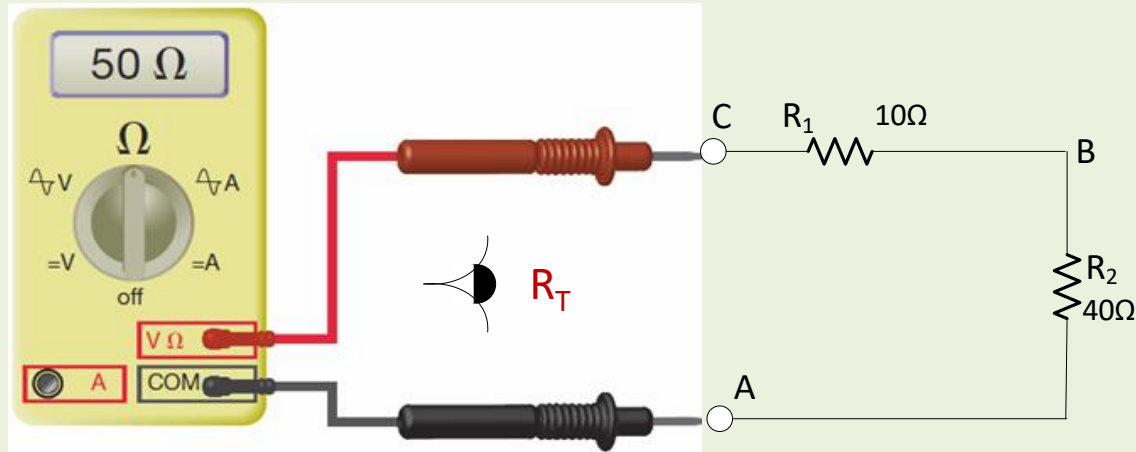
Question: How do we relate I_T , I_{R1} and I_{R2} ?

$$I_T = I_{R1} = I_{R2}$$



Refer to Smartbook Sec 8.6 Series Connection of resistors

The **total resistance**, R_T , of a series circuit is equal to the **sum** of the individual load resistances.

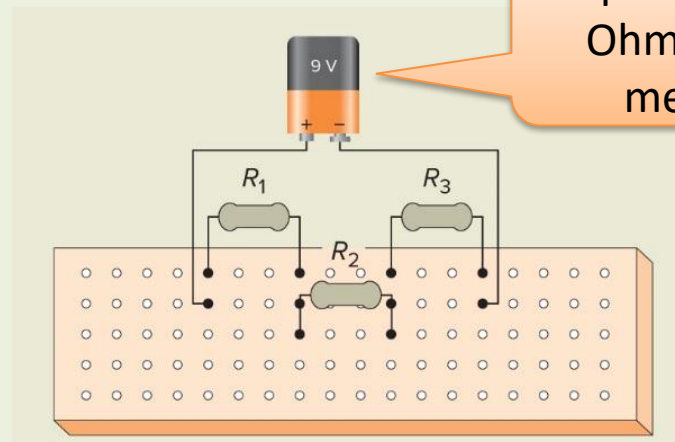


Why are there no voltage polarities shown on the schematic?

Question: $R_1(5k\Omega)$, $R_2(3k\Omega)$ and $R_3(2k\Omega)$ are connected in series. Find R_T

$$R_T = \boxed{50} \Omega$$

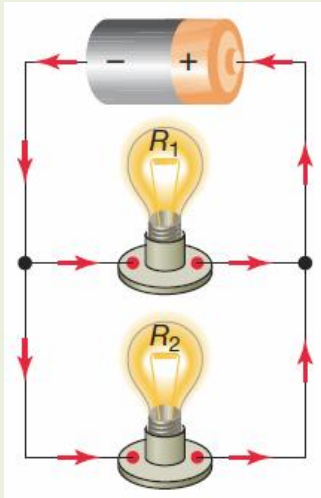
Replace battery with Ohmmeter when measuring R_T



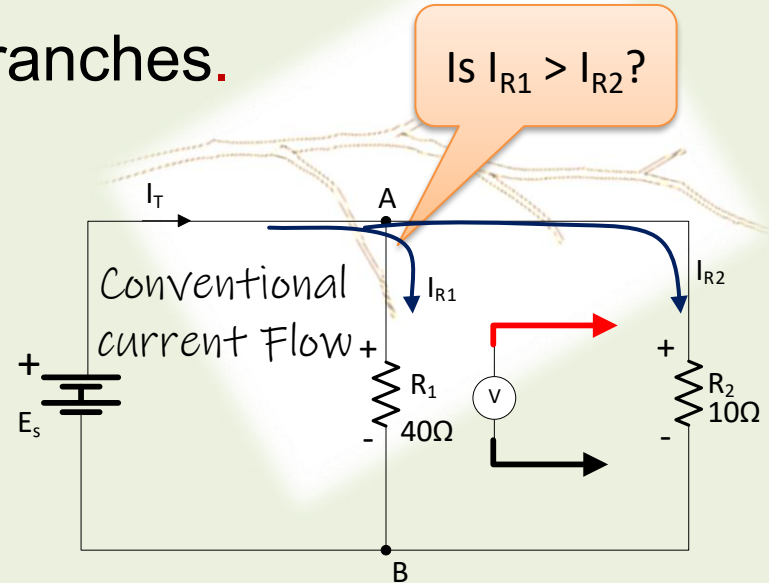


Refer to Smartbook Sec 12.1 Parallel Circuit Connection

A **parallel circuit** has **two or more paths** for current flow.
The loads connect **across** each other.
Total current **splits** to different branches.



Electron Flow



Is $E_s = 10V$, What is the value of V_{R2} ?

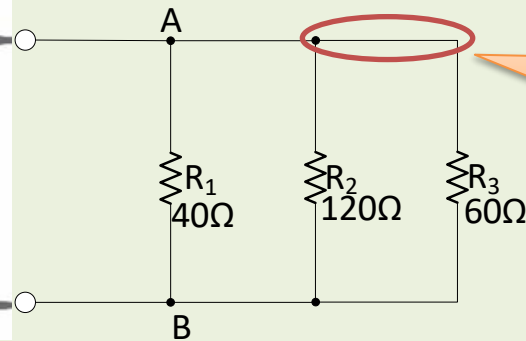
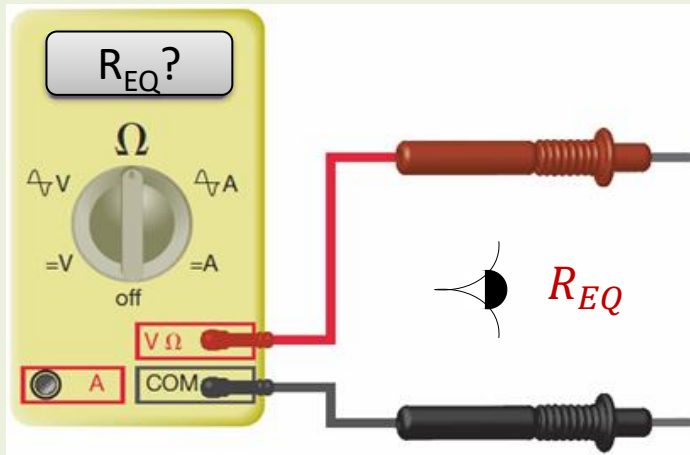
Question: How do we relate I_T , I_{R1} and I_{R2} ?

$$I_T = I_{R1} + I_{R2}$$



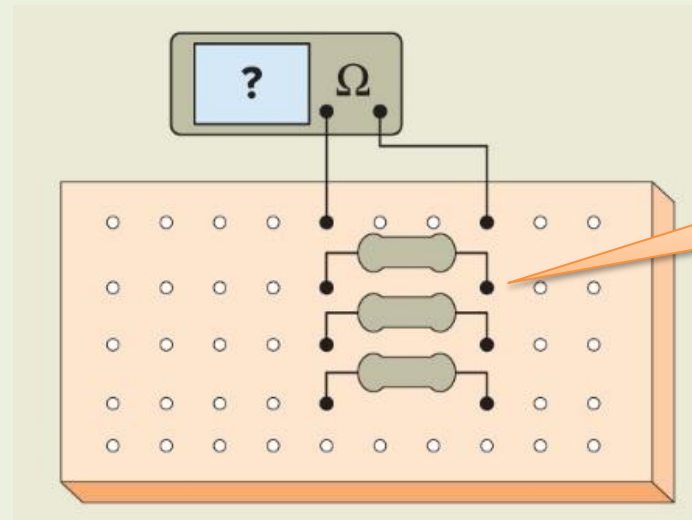
Refer to Smartbook Sec 8.6 Parallel Connection of resistors

The **Equivalent resistance** of a parallel circuit, $R_{EQ} = \frac{1}{\frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}}$



How shall we name these node?

$$R_{EQ} = \boxed{20} \Omega$$



Can we put R_1 on top of R_2 ?



Refer to Smartbook Chapter 7.1 Metric Prefixes

- Electrical quantities are typically very large or very small
- Engineering notation specifies that all powers of ten must be in the multiples of 3 or zero

- $1000\text{m} = 1 \times 10^3 \text{ m}$
- $1\text{m} = 1 \times 10^0 \text{ m}$

- Engineering notation can be replaced with corresponding metric prefix

- $1000\text{m} = 1 \times 10^3 \text{ m} = 1 \text{ kilo-meter}$

$$3000000\text{Hz} = 3 \times 10^6 = 3\text{MHz}$$

Decimal point Left
Shift

Metric Prefix	Symbol	Numbers	Engineering Notation
Tera	T	1,000,000,000,000	10^{12}
Giga	G	1,000,000,000	10^9
Mega	M	1, 000,000	10^6
Kilo	K	1,000	10^3
Milli	m	0.001	10^{-3}
Micro	μ	0.000001	10^{-6}
Nano	n	0.000000001	10^{-9}
Pico	P	0.000000000001	10^{-12}



Refer to Smartbook Chapter 7.1 Metric Prefixes

Voltage	SI Unit	Unit for very small amounts		Unit for very large amounts
Symbol	V	μ V	mV	kV
Pronounced	Volt	microvolt	millivolt	kilovolt
Multiplier	1	0.000001	0.001	1,000
Everyday Life		Noise voltage	Diode/LED voltage	Electrical Substation

Resistance	SI Unit	Unit for very small amounts	Unit for very large amounts	
Symbol	Ω	m Ω	k Ω	M Ω
Pronounced	Ohm (Ω)	milliohm	kiloohm	Megaohm
Multiplier	1	0.001	1,000	1,000,000
Everyday Life		Resistance of wire	Common resistor value	Body resistance

What is the common current (Ampere) value found in everyday life?

Current	SI Unit	Unit for very small amounts		Unit for very large amounts	
Symbol	A	μ A	mA	A	
Pronounced	Ampere	Microampere	milliampere	Ampere	
Multiplier	1	0.000001	0.001	1	
Everyday Life		Noise current	Typical current reading	Current limit for AC plug is 13A	



Refer to Smartbook Sec 5.6 Computer Simulation of Circuits

- Computer simulation software** allows you to experiment with electrical and electronic circuits without using laboratory facilities and materials.

Installation Guide



The screenshot displays the Micro-Cap 12.2.0.3 software interface. On the left, a file explorer window shows the directory structure: This PC > Data (D:) > MC12. The file 'mc12_64' is highlighted with a red circle and a red arrow pointing to it. The main window shows a circuit simulation with various components including switches (S1, S2, S3), resistors (R1, R2, R3, R4, R5, R6, R7, R8, R9, R10, R11), capacitors (C1, C2, C3, C4, C5, C6, C7, C8, C9, C10, C11, C12, C13, C14, C15, C16, C17, C18, C19, C20, C21, C22, C23, C24, C25, C26, C27, C28, C29, C30, C31, C32, C33, C34, C35, C36, C37, C38, C39, C40, C41, C42, C43, C44, C45, C46, C47, C48, C49, C50, C51, C52, C53, C54, C55, C56, C57, C58, C59, C60, C61, C62, C63, C64, C65, C66, C67, C68, C69, C70, C71, C72, C73, C74, C75, C76, C77, C78, C79, C80, C81, C82, C83, C84, C85, C86, C87, C88, C89, C90, C91, C92, C93, C94, C95, C96, C97, C98, C99, C100), and a motor (MOTOR1). The circuit is annotated with text: 'S1 and S2 are grouped as G1. Double-click on either and both change state on and off.', 'Double-click on S3 to see the motor speed and meter change.', and 'The use of animated analog DC analysis.' The software interface includes a menu bar (File, Edit, Component, Windows, Options, Analysis, Design, Model, Help), a toolbar, and a status bar at the bottom showing 'Page 1 / Text / Inf / 1'.

Click on this to run Micro Cap 12

Learning Objectives



- Compare the difference between Electron flow and Conventional Current Flow.
- Compare the current directions and voltage polarities in a DC circuit that contains an electrical source that delivers power and the load that consumes power.
- Apply Ohm's Law Triangle to determine three electrical quantities (Voltage, Current, and Resistance)
- Distinguish the application of using a Digital Multimeter to measure three electrical quantities.
- Identify the importance of appropriate use of metric prefix of values, such as milli (m) (10^{-3}), Kilo (k) (10^3), Mega (M) (10^6), etc.
- Categorize the common International System (SI) units use in electrical circuit quantities.

CA1- Lesson 02

Series and Parallel Circuits

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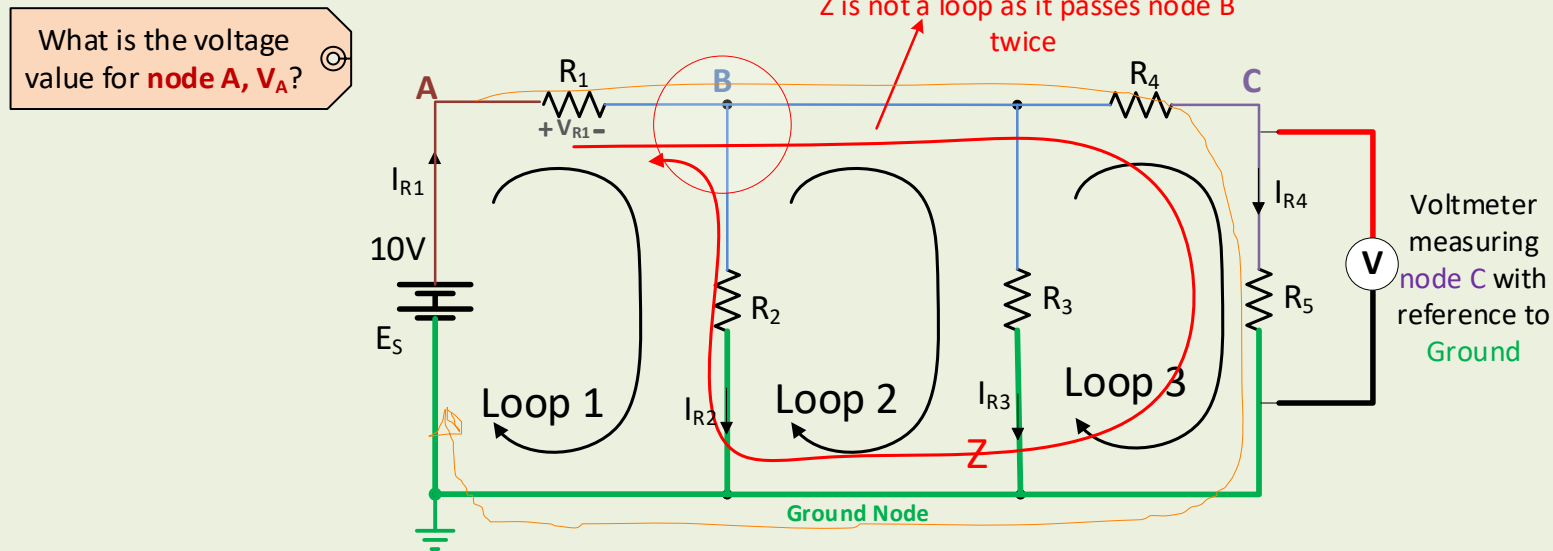
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Principles of
Electrical and
Electronic
Engineering



A Basic Circuit

- A loop in an electrical circuit is a closed path that starts at a node, proceeds through some circuit elements(branches) , and returns to the starting node.
 - A loop cannot pass through any node more than once
 - Loop X & Y is a valid loop but not **Z** which **passes by Node B twice**



Question:

V_{AB} = Voltage difference between **node A** and **node B**

$$= V_A - V_B$$

$$= V_{R?}$$

Find out the resistor voltage for:

$$V_B = V_{R_}$$

$$V_B - V_C = V_{R_}$$



Refer to Smartbook Sec 13.3 Kirchhoff's Voltage Law

Kirchhoff's Voltage Law (KVL): the **sum of all** the voltage drops **in series is equal** to the applied voltage.

$$\text{KVL: } E_T = V_{R1} + V_{R2} + V_{R3}$$

Ohm's law for each resistor voltage:

$$V_{R2} = I_T \times R_2,$$

Hence, KVL can be rewritten as below,

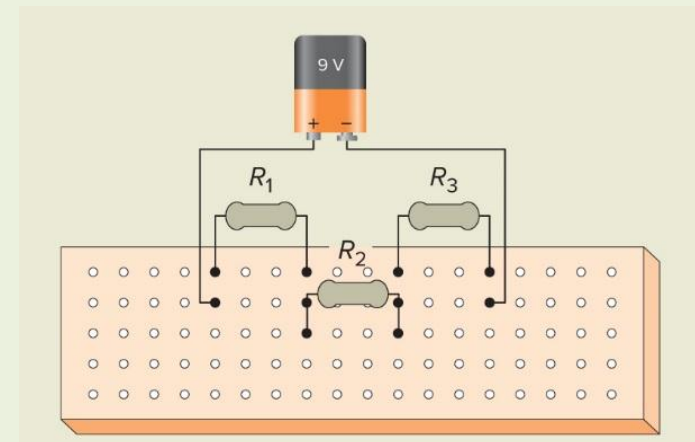
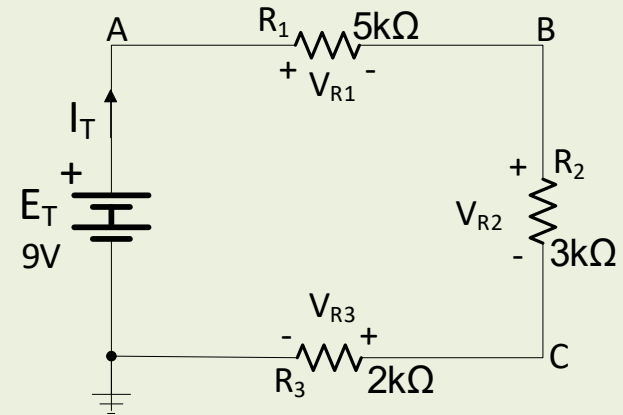
$$\text{KVL: } E_T = (I_T \times R_1) + (I_T \times R_2) + (I_T \times R_3)$$

$$E_T = I_T \times (R_1 + R_2 + R_3)$$

Question: Apply KVL to find I_T

$I_T =$

mA





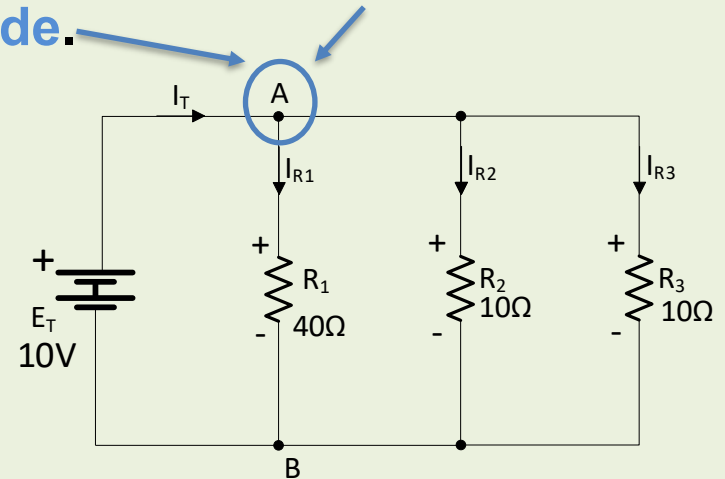
Refer to Smartbook Sec 13.2 Kirchhoff's Current Law

Kirchhoff's Current Law (KCL): **current entering** any **Node** in a circuit is equal to **current leaving** that same **Node**.

This relationship is expressed as

$$KCL \text{ @ Node A: } I_T = I_{R1} + I_{R2} + I_{R3}$$

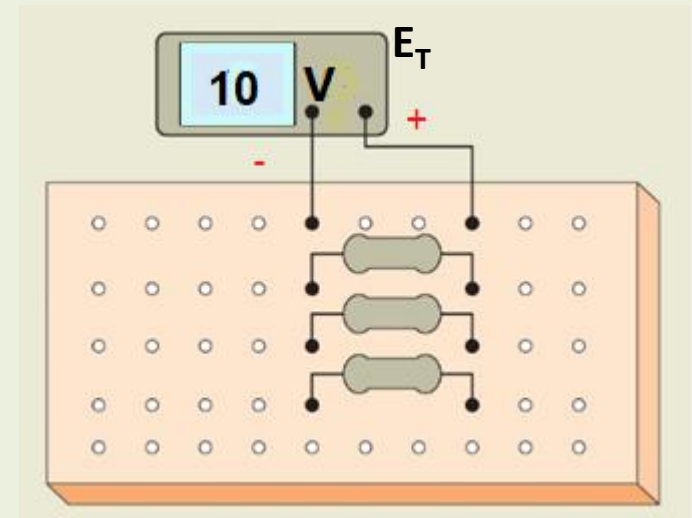
KCL is very useful to analyze **parallel circuit**.



Question: Apply KCL to find I_T

$$I_T = \boxed{} \text{ mA}$$

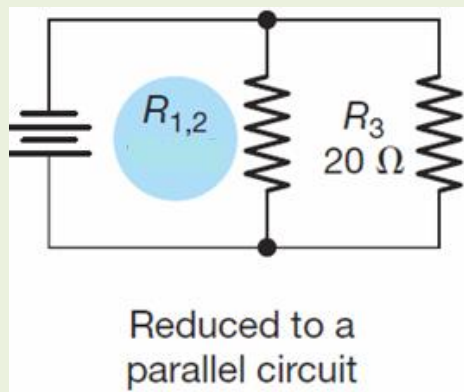
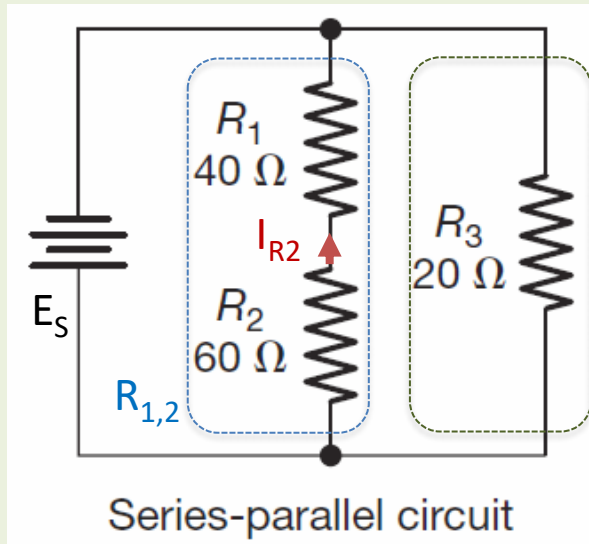
Can we apply KCL at Node B?





Refer to Smartbook Sec 13.4 Determining Resistance

To the **source**, any series-parallel resistive network can be simplified to a **single resistor**, R_{EQ} .



Analysis of series-parallel resistive network

E_S saw 2 parallel branches, and

is the combination of 2 series resistors can be simplified to 1 single resistor, $R_{1,2} =$

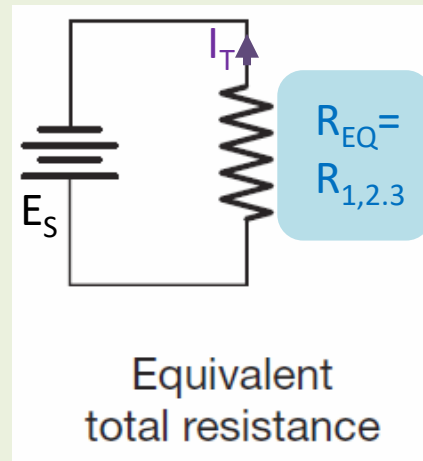
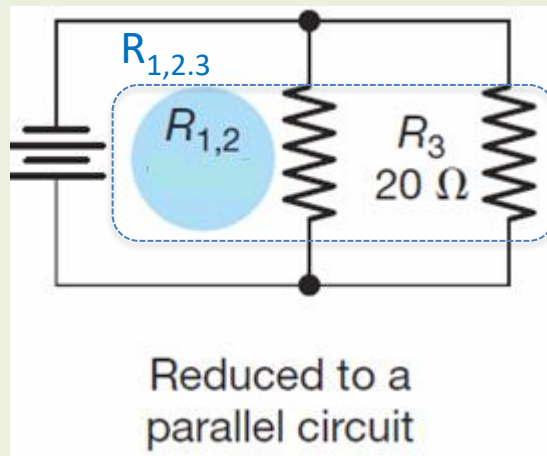
$$R_{12} = (\quad) \Omega$$

$$\text{Hence, } V_{R12} = V_{R3} = E_S$$



Refer to Smartbook Sec 13.4 Determining Resistance

To the **source**, any series-parallel resistive network can be simplified to a single resistor, R_{EQ} .



$$R_{123} = \quad \Omega$$

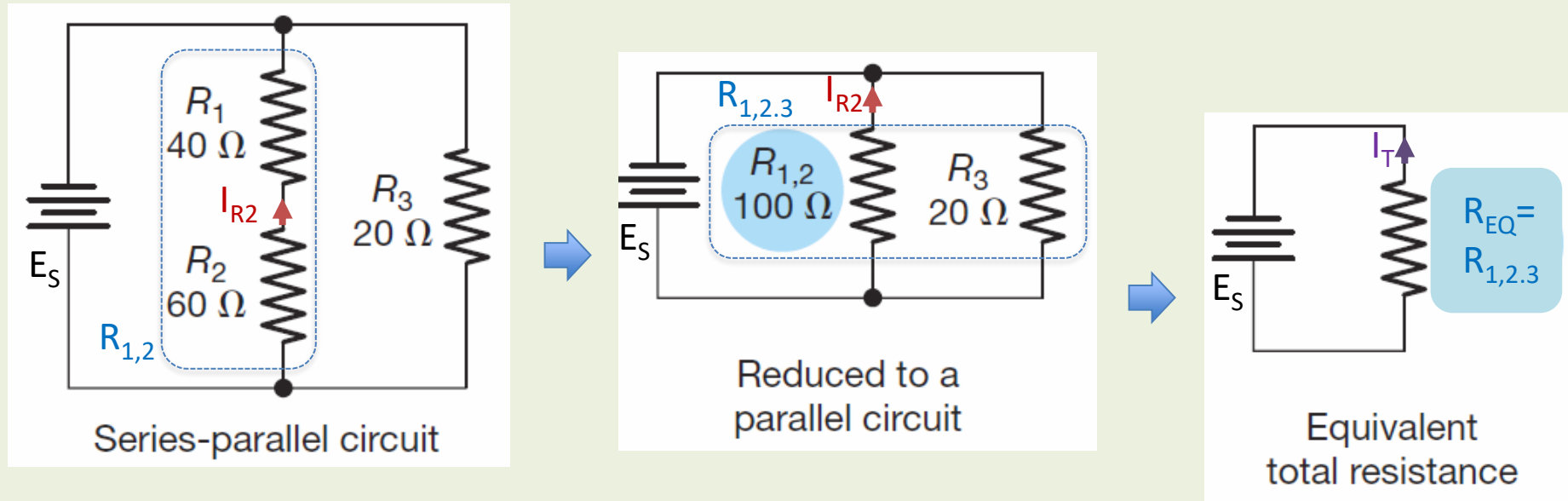
Based on the calculated R_{EQ} to determine the source current, I_T if $E_S = 5V$.

$$I_T = \quad mA$$



Refer to Smartbook Sec 13.4 Determining Resistance

To the **source**, any series-parallel resistive network can be simplified to a single resistor, R_{EQ} .



Based on the information gathered in Parts 1 and 2, apply KCL to determine the values of the 2 branch currents, I_{R2} and I_{R3}

$$KCL: I_T = I_{R2} + I_{R3}$$

$$I_{R2} = 50\text{mA}$$

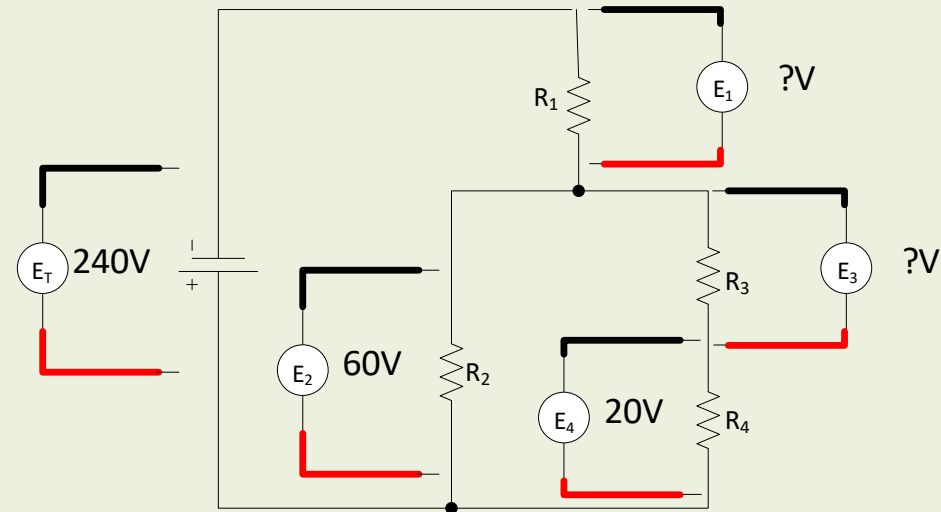
$$I_{R3} = 250\text{mA}$$



Refer to Smartbook Figure 13.8 Series-Parallel Circuit

Use KVL to solve the unknown voltage drops

- Step 1: Identify how the current flow in the circuit
- Step 2: Write down the voltage polarity on each resistor
- Step 3: Distinguish the series loop in the circuit to apply KVL



Question: Apply KVL to find resistor voltages

$$E_1 = \boxed{} V$$

$$E_3 = \boxed{} V$$



Refer to Smartbook Figure 13.5 Series-Parallel Circuit

Use KCL to solve the unknown currents

Step 1: Identify the direction of all branch currents in the circuit

Step 2: Label all the node names in the circuit

Step 3: Equate the current in and current out of every node to form KCL equation

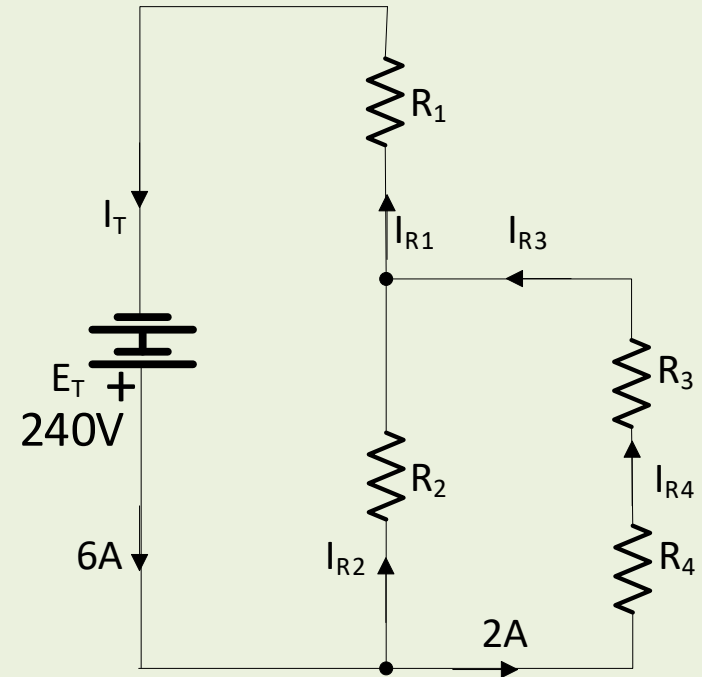
Question: Apply KCL to find the value of the following branch currents

$$I_{R2} = \boxed{} \text{ A}$$

$$I_{R1} = \boxed{} \text{ A}$$

$$I_{R3} = \boxed{} \text{ A}$$

$$I_T = \boxed{} \text{ A}$$





Refer to Smartbook Sec 13.2 Kirchhoff's Current Law

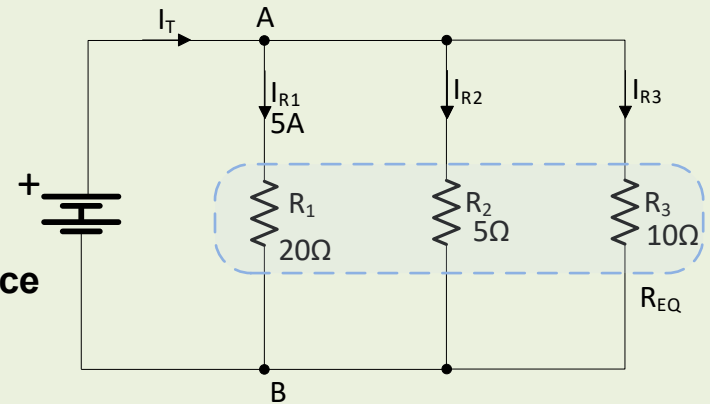
Parallel resistor circuits act as **current dividers** as the **current splits** into different branches with current value depend on the branch resistance.

$$I_{RX} = \frac{R_{EQ}}{R_X} \times I_T$$

Where:

I_S = the total current

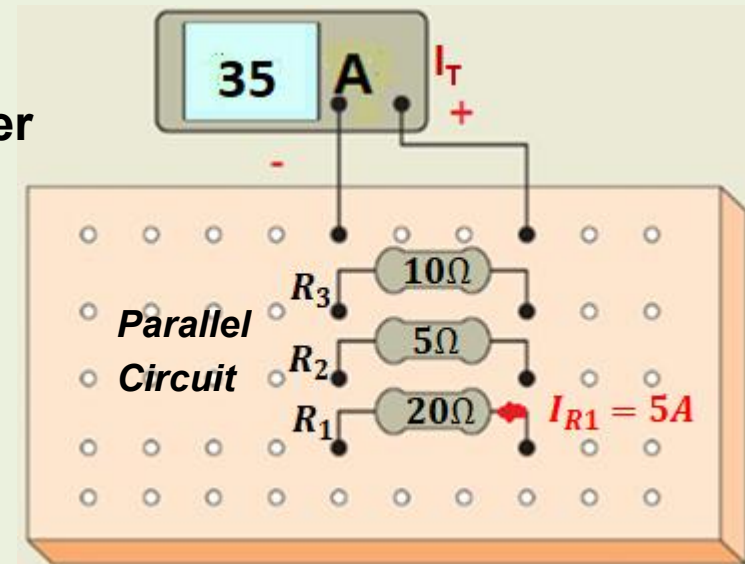
R_{EQ} = the **equivalent** parallel resistance



Question: Given that $I_T = 35A$, apply **Current Divider** to verify $I_{R1} = 5A$

$$R_{EQ} = R_1 // R_2 // R_3 = \quad \Omega$$

$$I_{R1} = \frac{R_{EQ}}{20} \times 35 = 5A$$



Learning Objectives



- Contrast series and parallel connections in terms of the three electrical quantities in any DC electrical circuit.
- Calculate Equivalent resistance, R_{EQ} of a Series-Parallel resistive network.
- Apply Kirchhoff's Voltage Laws (KVL) in series circuit.
- Apply Kirchhoff's Current Laws (KCL) in parallel circuit.